

AI BASED INTEGRATED SOLUTION FOR SMART PEPPER FARMING IN SRI LANKA

25-26J-479

Project Proposal Report

Adhikari A. A. J. T.

B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidate is carrying out research for the undergraduate dissertation under my supervision.

 Signature of the supervisor (Ms. Ishara Weerathunga)	<u>23/08/2025</u> Date
 Signature of the co-supervisor (Mr. Eishan Weerasinghe)	<u>24.08.2025</u> Date

ABSTRACT

Sri Lanka is one of the leading exporters of pepper, and its Ceylon pepper is well known for high piperine content and premium quality. However, most smallholder farmers still rely on manual and traditional methods to evaluate post-harvest pepper quality. These methods are often subjective, time-consuming, and not aligned with international standards, which creates difficulties in gaining better market prices and export opportunities. This research proposes the development of an Artificial Intelligence (AI), and Internet of Things (IoT) powered quality grading system to provide farmers with an affordable, reliable, and transparent solution.

The system is designed around three main components. First, an IoT-based bulk density device using an ESP32 microcontroller and load cell will replace manual cup methods, giving farmers a more accurate and consistent measurement. Second, an AI-based image analysis module will process photos taken by farmers using their mobile phones to detect defects such as mold, insect damage, berry size variation, and adulteration. Third, the system will include features to estimate piperine levels based on pepper variety and location, while also supporting Good Agriculture Practices (GAP) certification validation to improve trust with buyers.

A mobile application will guide farmers step by step, provide instructions with images or short videos, and generate a digital grading report that can be shared with buyers. The report will include grading results, piperine estimation, and packaging recommendations based on export standards. In addition, the app will keep a history of results and offer feedback to help farmers improve future harvests.

This research aims to reduce subjectivity in pepper grading, empower farmers with accessible technology, and strengthen Sri Lanka's competitive advantage in the global spice market. The outcome will support both farmers and exporters by building confidence, ensuring transparency, and promoting sustainable farming practices.

Key Words - Pepper Quality Grading, IoT Device, Bulk Density, Computer Vision, AI Image Analysis, GAP Certification

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
GAP	Good Agricultural Practices
IoT	Internet of Things
IPC	International Pepper Community
ISO	International Organization for Standardization
PDF	Portable Document Format
QR	Quick Response
RAM	Random Access Memory
SVM	Support Vector Machine

Sub-Topic: Smart Pepper Quality Evaluation Using Image Analysis, IoT Bulk Density Measurement, and Certificate Validation

1. INTRODUCTION

Sri Lanka is one of the major producers and exporters of black and white pepper, which is an important spice in both local and international markets. However, the pepper industry in the country faces many challenges, especially in improving productivity, ensuring quality, and meeting global market standards. Most smallholder farmers still rely on traditional methods, with limited access to modern digital tools that can guide their decision-making. Because of this, Sri Lankan pepper often struggles to compete with other exporters such as Vietnam, India, and Indonesia, even though Ceylon pepper is known for its superior piperine content and unique flavor profile [1], [2], [11].

To address these issues, our research proposes an integrated Artificial Intelligence (AI), and Internet of Things (IoT) powered intelligent decision support system for smart pepper farming in Sri Lanka. This system will combine multiple modules such as yield prediction, disease detection, market price forecasting, post-harvest quality grading, and blockchain-based traceability. Together, these modules will help farmers make better decisions, increase transparency, and improve both profitability and sustainability in the pepper value chain [12], [13].

Within this system, my component focuses on post-harvest quality grading. At present, most Sri Lankan farmers grade pepper using a 1L cup to check bulk density (e.g., 550g/L, 500g/L, 450g/L). While this method is simple and low-cost, it is inconsistent and considers only one parameter. It ignores other critical factors such as mold, insect damage, adulteration (e.g., papaya seeds), berry size, and color uniformity. International standards such as ISO 959 and Codex CXS 326-2017 require grading to consider density, cleanliness, size, color, and contaminants [1] - [3]. Therefore, the current farmer-level

practices do not align with International Organization for Standardization (ISO) and International Pepper Community (IPC) requirements [4].

Several studies have attempted to improve spice grading. For example, the iPepper system combined image features with moisture data to check pepper quality [8], while PepperTester introduced a non-destructive method for purity testing [6]. Other works applied machine learning to classify pepper seeds [7] or to detect adulteration using image analysis. These show the potential of AI and computer vision for spice grading. However, most of these systems remain limited in scope, lab-based, or not practical for smallholder farmers.

At present, there is still no integrated solution that combines:

- Physical checks such as bulk density (using IoT devices),
- Visual grading of mold, insect damage, light berries, adulteration, and color (using AI image analysis), and
- Certification validation such as Good Agriculture Practices (GAP), which is important for gaining buyer trust and export compliance [16].

My proposed system will bring these together into a farmer-friendly mobile application. Farmers will connect a low-cost IoT device (ESP32 + load cell) via Bluetooth to measure bulk density, capture images of pepper samples using their phone, and validate GAP certification through the app. The system will then generate a digital grading report (PDF) showing the final grade, quality breakdown, and packaging guidance. Farmers can also track their grading history and receive recommendations to improve drying, sorting, and storage practices. Additionally, premium-quality batches will trigger notifications to buyers, strengthening the link between farmers and the export market.

This approach is different from previous studies because it moves beyond single-parameter grading. By combining IoT, AI, and certification validation, the system creates

a multi-parameter, farmer-friendly, and export-compliant grading framework. This directly supports Sri Lanka's pepper industry by ensuring fair value for farmers, building trust with buyers, and promoting the global recognition of Ceylon pepper.

1.1 Background and Literature Survey

Pepper (*Piper nigrum L.*) is one of the most valuable spices grown in Sri Lanka, exported mainly as black and white pepper. The crop is a key income source for smallholder farmers, yet the industry faces challenges in maintaining uniform and high-quality standards. Traditionally, grading at farm level is carried out using bulk density, where a one-liter cup is filled and weighed to estimate grams per liter (g/L). While density is an important parameter, this method is often error-prone, can be manipulated, and ignores other crucial factors such as berry color, size, mold, insect damage, and adulteration. As a result, there is a clear mismatch between farmer practices and global standards.

International benchmarks such as ISO 959 [1], [2], Codex Alimentarius [3], and IPC guidelines [4] define multiple grading factors. These include bulk density, color uniformity, cleanliness, extraneous matter, and the absence of mold or adulteration. In Sri Lanka, the Department of Export Agriculture has also highlighted differences among local pepper varieties and reported wide variation in piperine content [5]. For example, Ceylon pepper is known to contain 7-15% piperine, a much higher level than some hybrid varieties. This makes Ceylon pepper unique in international markets but also increases the need for accurate testing and transparent grading. GAP certification is another important factor, as it provides assurance of good agricultural practices and improves buyer confidence in Sri Lankan exports [16].

Several studies have tried to address the weaknesses of manual grading. PepperTester introduced a non-destructive method to test purity of black pepper seed samples, but it mainly focused on seed authenticity rather than full grading [6]. The iPepper system

combined image processing with density and moisture data to automate quality checks. While it was a step forward, it was limited in scope and not farmer friendly [8]. More recent studies have used machine learning and feature extraction to identify pepper quality based on visual cues, reaching high accuracy for defect detection [7]. A recent study on cardamom grading using YOLOv8 achieved 92% accuracy in detecting and classifying based on size, shape, and texture [17]. Another study classified pepper varieties using Support Vector Machine (SVM) with 87% accuracy on more than 5,000 samples [18]. These results show that AI-driven grading methods are both feasible and effective for spices.

Piperine is the compound responsible for pepper's pungency and market value. Classic laboratory methods such as chromatography [9] and newer studies [10] provide accurate measurements, but these techniques require expensive equipment and trained staff, making them inaccessible to small-scale farmers. Research on Sri Lankan pepper varieties confirms that piperine levels differ significantly across cultivars [11]. This highlights the importance of creating low-cost field-level estimation methods or indirect grading systems that can reflect this variation.

Another concern in the pepper trade is adulteration. Papaya seeds and other fillers are sometimes mixed with pepper to increase weight. ISO and IPC standards require strict rejection of such adulterated batches, but manual farmer-level grading methods cannot detect them reliably. Mold detection is equally critical; even 1-2% mold berries can lower an entire batch's grade from premium to standard. Advanced image recognition techniques have been tested in other crops such as coffee beans [15] and seed grading [14], showing the potential of AI-driven approaches for pepper as well.

In agriculture more broadly, IoT devices using load cells have been applied for automated grading of grains, fruits, and spices [12], [13]. Computer vision models such as MobileNet and YOLO have achieved strong accuracy in defect detection across crops. In Sri Lanka,

researchers have also begun exploring AI-based disease prediction for pepper and blockchain for transparency in spice supply chains. These are important advances, but they remain isolated solutions, not integrated systems for holistic grading.

In summary, most existing research addresses only one aspect whether density, purity, image-based classification, or chemical analysis. PepperTester [6] focused mainly on seed purity, while iPepper [8] combined images with moisture data but was not farmer friendly. The Pepper Berries Graded with Deep Learning Models study [19] showed strong results in controlled conditions but was not tested in real-world farms. Similarly, the Pepper Seed Classification using SVM [18] proved the potential of variety-based grading, while the Cardamom Grading study with YOLOv8 [17] highlighted the success of AI in spice grading. However, none of these systems integrate physical measurements, visual analysis, piperine estimation, and certification into one platform. Therefore, this research aims to fill that gap by proposing a multi-parameter, farmer-friendly grading module that combines IoT-based density measurement, AI-driven image analysis, piperine estimation, and GAP certification checks. By aligning with IPC and ISO standards while keeping the solution practical for smallholder farmers, this project builds on past research while introducing a new, integrated, and scalable approach for the Sri Lankan pepper industry.

1.2 Research Gap

Many studies have tried to improve the grading and quality assessment of pepper, but most of them are either limited in scope or not suitable for smallholder farmers. In Sri Lanka, farmer-level grading is still mainly based on bulk density, using a one-liter cup to categorize pepper into grades such as 550g/L, 500g/L, and 450g/L. Although this method is simple and low-cost, it is error-prone, inconsistent, and ignores other important quality factors such as mold, adulteration, extraneous matter, or color uniformity. This shows a clear gap between local practices and the comprehensive grading requirements outlined by ISO and IPC standards [1] - [4].

PepperTester introduced a non-destructive approach to test the purity of black pepper seed samples [6]. While it proved that automated analysis is possible, the study mainly focused on authenticity and purity. It did not consider multi-parameter grading such as moisture, mold, or extraneous matter.

iPepper system used image features combined with moisture data for quality checks [8]. This was a step forward, but it was limited in scope and lacked a farmer-friendly interface. It also did not integrate physical, visual, and certification-based grading in one system.

Black Pepper Grading using Feature Extraction and Machine Learning showed that visual features such as color and shape can classify pepper berries with good accuracy [7]. However, the work was tested only in controlled experiments and not scaled to real farmer-level use. Similarly, the study Pepper Berries Graded using Deep Learning Models (ICCES 2024) applied advanced neural networks for multi-class grading of pepper berries [19], but this approach was tested on datasets rather than field conditions, limiting its direct usability for farmers.

The Classification of Pepper Seeds using SVM with Color Filter Array Images research achieved around 87% accuracy in variety classification [18]. This work supports the idea of variety-based grading and indirect piperine estimation but does not cover other quality parameters like mold, density, or adulteration. Likewise, the Cardamom Grading study using YOLOv8 reached 92% accuracy in detecting defects [17], demonstrating the potential of AI in spice grading, but it was crop-specific and not applied to pepper.

Research on piperine, the compound responsible for pepper's pungency and value, has also highlighted differences between Sri Lankan pepper and hybrids [9] - [11]. While lab-based chromatography methods provide accurate results, they are expensive and not

practical for farmers. As a result, there is still no field-level system that can help farmers estimate or benefit from this unique advantage of Ceylon pepper.

Other trade issues such as adulteration (e.g., papaya seeds mixed with pepper), and mold detection remain major challenges. Even a small percentage of mold berries can reduce a batch from premium to standard grade. Current farmer-level practices cannot detect these issues reliably. Studies in other crops, such as coffee beans [15] and seed grading [14], show that AI-based image recognition can successfully detect defects, which suggests strong potential for pepper as well.

Beyond pepper, many IoT and AI applications in agriculture show the potential for scalable solutions. For example, IoT frameworks using ESP32 with load cells have been successfully applied to monitor crop quality and environmental conditions [12], [13]. Mobile-based computer vision models such as MobileNetV2 have achieved real-time quality assessment for seeds and coffee beans [14], [15]. These studies prove that IoT and mobile vision can deliver accurate, low-cost, and real-time grading. However, such systems are mostly applied to crops like rice, tea, or coffee, and very few target spices like pepper.

From this comparison, it is clear that existing research tends to address only one factor at a time, such as density, purity, image-based classification, or chemical testing. None of them combine physical (bulk density), visual (mold, adulteration, color, texture), and documentation-based (certification) parameters into a unified grading framework. In particular, GAP certification, which is promoted by the Department of Export Agriculture in Sri Lanka to ensure quality and sustainability [16], has not been integrated into any previous grading system.

Therefore, the main research gap lies in the absence of an integrated, farmer-friendly system that:

1. Uses an IoT device (ESP32 + load cell) for bulk density measurement.
2. Employs AI-driven mobile image analysis for visual defects like mold, adulteration, or uneven coloration.
3. Incorporates GAP certification validation to ensure compliance with global standards.

The proposed system aims to bridge this gap. Unlike existing approaches, it introduces a multi-parameter grading framework aligned with ISO and IPC standards, while also providing transparency features such as digital grading reports, historical feedback for farmers, and buyer notifications for premium-quality batches. This novelty makes the solution both technically innovative and practically valuable for smallholder farmers in Sri Lanka.

Table 1.1: Research Gap

Research / Method	Bulk Density	Visual Defects	Piperine (Chemical)	Certification	Multi-Parameter	Farmer-Friendly
Traditional 1L Cup Method	✓	✗	✗	✗	✗	✓ (simple, but inaccurate)
ISO 959 / IPC Standards [1 - 4]	✓ (required)	✓ (required)	✓ (lab only)	✓	✗	✗ (benchmarks, not tools)
PepperTester (2025) [6]	✗	✓ (purity only)	✗	✗	✗	✗
Piperine Studies [9 - 11]	✗	✗	✓ (lab-based only)	✗	✗	✗

iPepper (2011) [8]	✗	✓ (limited , image + moisture)	✗	✗	✗	✗
Black Pepper Grading (ML Features) [7]	✗	✓ (color, shape classification)	✗	✗	✗	✗ (tested only in lab)
Pepper Berries with Deep Learning [19]	✗	✓ (multi-class grading , datasets)	✗	✗	✗	✗ (not tested in field)
AI/IoT in Other Crops [12 - 15]	✓ (IoT devices used)	✓ (AI defect detection)	✗	✗	✗ (partial, isolated use)	✗ (limited, not pepper-specific)
Proposed Research	✓ (IoT: bulk density)	✓ (AI: mold, color etc.)	✓ (variety-based estimation)	✓ (GAP certification)	✓ (fully integrated)	✓ (mobile app with reports and feedback)

1.3 Research Problem

The main research problem in this study is how to accurately and transparently evaluate the quality of black and white pepper at the farmer level, in a way that matches international standards but is still practical and affordable for smallholders.

At present, most farmers in Sri Lanka grade pepper using only bulk density, measured with a 1L cup. This method is simple, but it is also error-prone because of inconsistent weighing, inaccurate cup sizes, and personal judgment. More importantly, it considers only one factor while ignoring other critical aspects such as mold, adulteration, berry size, and GAP certification status.

Because of this, farmers often fail to prove the true quality of their harvest. Buyers and exporters face mismatches between farmer-graded batches and standards such as ISO 959 and IPC specifications [1] - [4]. This often leads to rejected consignments, reduced trust, and Sri Lankan pepper losing its premium position in global markets. Farmers also lose fair income when premium-quality produce is downgraded by weak grading practices.

Several studies have tried to solve parts of this issue. For example, PepperTester focused only on purity testing [6]. iPepper [8] combined image features and moisture data but was not farmer friendly. Black Pepper Grading with Machine Learning [7] and Pepper Berries Graded using Deep Learning Models [19] showed that AI can classify pepper visually, but both were tested on datasets, not real farm-level conditions. Other works on piperine [9] - [11] confirmed differences between varieties, but lab tools like chromatography are too costly for farmers. IoT frameworks using ESP32 with load cells [12], [13] have shown success in other crops, but not yet in pepper.

Therefore, the real problem is that there is no integrated and farmer-friendly solution that can:

1. Measure bulk density using a simple IoT device.
2. Detect visual defects such as mold, adulteration, and foreign matter using AI image analysis.
3. Validate certifications such as GAP [16], which are promoted by the Department of Export Agriculture in Sri Lanka.

4. Provide farmers with clear digital reports, historical feedback, and improvement tips.
5. Guide farmers with standard packaging instructions, so their produce remains compliant for export.
6. Notify buyers when premium-quality batches are available.

In simple terms, the problem is that existing grading practices and research fail to give farmers a complete, practical, and transparent way to evaluate pepper quality. This research addresses that gap by developing a low-cost, mobile-based solution that combines IoT and AI to ensure accurate grading, empower farmers with knowledge, and build stronger trust between farmers, buyers, and global markets.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to develop an integrated AI- and IoT-powered intelligent decision support system for smart pepper farming in Sri Lanka. The proposed system will combine several key functions such as yield prediction, disease detection, market price forecasting, post-harvest quality grading, and blockchain-based traceability into a single farmer-friendly platform. By integrating these areas, the research aims to improve productivity, transparency, and profitability for smallholder farmers, while strengthening the position of Sri Lankan pepper in international markets.

The first component will focus on yield prediction. Farmers often rely on guesswork to estimate harvests. This research proposes to train machine learning models using plant growth and environmental data. Augmented reality (AR) will also be explored to visualize yield per plant and field growth patterns, helping farmers plan resources more effectively.

The second component will address disease detection. Since pepper crops are highly vulnerable to fungal and bacterial infections, an AI-based mobile tool will be developed to detect early signs of disease from leaf images. This will provide timely alerts and treatment recommendations, reducing crop loss and supporting sustainable practices.

The third component is market price forecasting. Pepper prices are highly volatile, leading to poor selling decisions. The system will use AI forecasting models trained on historical price trends, seasonal data, and international trade information to predict price movements. Farmers will then receive advice on the most profitable times to sell.

The fourth component is post-harvest quality grading. Current grading in Sri Lanka relies mainly on bulk density, which is error-prone and incomplete. The proposed grading module will combine IoT-based bulk density measurement with AI-driven visual defect

detection (mold, adulteration, color variation) and GAP certification checks. Packaging guidance will also be included to ensure export compliance.

Finally, the system will include a blockchain-based traceability feature. Each pepper batch will have a secure digital record of its origin, grading results, and certification status. Buyers and exporters will be able to scan Quick Response (QR) codes to verify these details, ensuring transparency and reducing fraud in the supply chain.

In summary, this research proposes a holistic digital solution that reduces the gap between traditional practices and modern technology. By integrating AI, IoT, AR, and blockchain into one platform, the system will empower farmers with reliable insights, fair grading, and better market opportunities. This will directly benefit smallholder farmers and enhance the long-term sustainability and global reputation of Ceylon pepper.

2.2 Specific Objectives

- **To develop an IoT-based device for bulk density measurement**

To design and implement a low-cost device using ESP32 and a load cell to measure the bulk density of pepper. This will replace the manual cup method and provide more accurate, consistent, and farmer-friendly results.

- **To apply AI-based image analysis for visual quality grading**

To use computer vision models (e.g., YOLOv8, MobileNetV3) to detect defects such as mold, insect damage, discoloration, berry size variation, and adulteration (e.g., papaya seeds). Farmers will be able to capture images through their mobile phones without needing high-end cameras.

- **To estimate piperine levels through variety and metadata**

To provide an indirect estimation of piperine content using variety, region, and

farmer input, since lab-based testing is not accessible to smallholders. This will highlight the natural advantage of Ceylon pepper, which has higher piperine levels compared to hybrids, and encourage farmers to grow more local varieties for better quality and market value.

- **To validate GAP certification**

To support GAP certification by linking grading results with certification status verified by the Department of Export Agriculture. The system will also raise awareness of the benefits of GAP, motivating farmers to obtain certification and improve buyer trust.

- **To provide user-friendly guidance and instructions**

To offer farmers step-by-step guides, images, and short videos for the grading process, making the system easy to use and building confidence among smallholders with limited technical knowledge.

- **To generate a digital grading report**

To create a simple and clear digital report (PDF) that summarizes grading results, piperine estimation, and certification status. Farmers will be able to share this report directly with buyers to increase transparency and trust.

- **To provide feedback and packaging recommendations**

To give farmers practical suggestions for improving quality in future harvests (e.g., drying methods, sorting techniques) and provide packaging instructions based on international standards, ensuring compliance with export requirements and maintaining quality during transport.

3. METHODOLOGY

3.1 System Overview

The proposed system is an integrated AI and IoT-powered intelligent decision support solution for smart pepper farming in Sri Lanka. The aim is to bring together different technologies such as computer vision, IoT sensors, machine learning, AR, and blockchain into one mobile-based platform that supports farmers and other stakeholders. The system is designed to improve yield estimation, disease management, price forecasting, post-harvest quality grading, and transparency in the supply chain.

- **Yield Prediction and Visualization**

This module will use machine learning models to predict the expected yield of pepper plants based on growth and environmental data. Farmers will be able to view yield information through a mobile app, supported with AR features to visualize yield at the plant level. Heatmaps will also be generated to show growth intensity and distribution across the field, helping farmers plan resources and make better management decisions.

- **Disease Detection and Health Monitoring**

This module will use AI-based image recognition to detect early signs of pepper diseases from simple mobile phone images. Farmers will receive timely alerts and treatment suggestions, which may include both organic and chemical options. The system will also support plant health monitoring to identify stress before visible damage occurs, reducing losses and improving crop management.

- **Market Price Forecasting and Blockchain Traceability**

This module will focus on predicting pepper prices using AI-based forecasting models trained with historical and seasonal data. Farmers will get “sell or wait” recommendations to maximize income. At the same time, blockchain technology will be used to store key information such as yield, disease status, quality grades, and certification details. A QR

code will be linked to each pepper batch, allowing buyers and exporters to verify the product's origin and quality history.

- **Post-Harvest Quality Grading**

This module will use an IoT device (ESP32 with load cell) to measure bulk density and combine it with AI image analysis for visual grading. The system will check for important factors such as berry size, color uniformity, mold, insect damage, and adulteration. Farmers will receive a digital grading report (PDF) with clear feedback and packaging instructions. Premium-quality batches will trigger buyer notifications, creating better market opportunities for farmers.

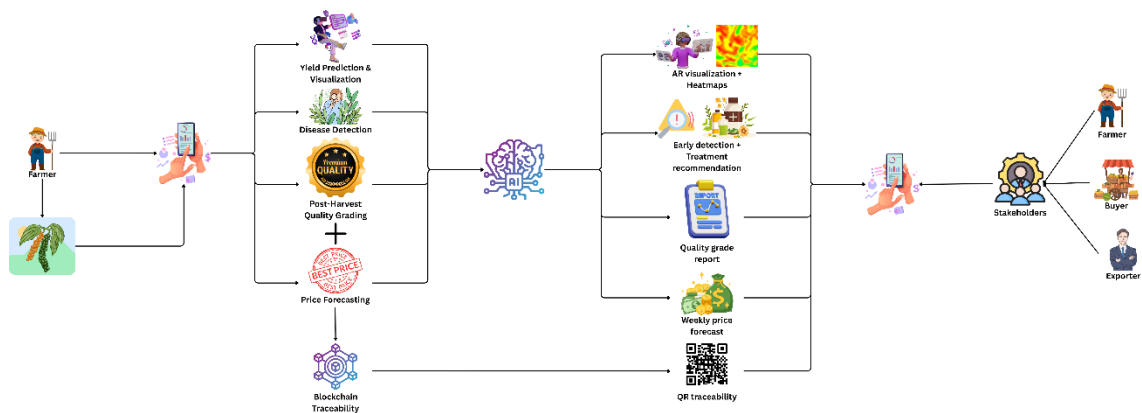


Figure 3.1: System Overview Diagram

3.2 System Architecture

The proposed system will be built as a modular AI and IoT-powered platform. It will connect farmers, devices, and cloud services to support all the major functions: yield prediction, disease detection, market price forecasting, post-harvest quality grading, and blockchain-based traceability. The design will follow a layered architecture to make the system scalable and easy to manage.

1. Data Acquisition Layer

This layer will handle the collection of raw data from farmers, IoT devices, and external sources.

- **IoT Devices and Sensors:** An ESP32-based device with a load cell will be used to measure bulk density of pepper samples.
- **Image Capture:** Farmers will use the mobile app to capture images of pepper cones (for yield prediction), leaves (for disease detection), and dried pepper samples (for quality grading).
- **External Data Sources:** Weather information, international price databases, and currency exchange rates will be connected for market price forecasting.

2. Data Processing and AI Analytics Layer

This layer will focus on analyzing the collected data using AI and machine learning models.

- **Yield Prediction Module:** Machine learning models will process pepper plant images to predict expected yield. AR and heatmaps will be used to visualize the results.
- **Disease Detection Module:** Deep learning will be used to identify disease symptoms early from leaf images, with treatment recommendations provided.
- **Market Price Forecasting Module:** AI-based forecasting (such as time-series models) will analyze historical and external data to predict future price trends and give “sell or wait” suggestions.
- **Post-Harvest Quality Grading Module:** IoT measurements (bulk density) will be combined with image analysis for defects like mold, insect damage, or adulteration. A grading score (e.g., Premium, Grade 1-3) will be assigned.
- **Blockchain Traceability:** Key outputs (yield, disease status, quality scores, certifications, and price forecasts) will be recorded on a secure blockchain system, with QR codes generated for each pepper batch.

3. Application and Visualization Layer

This layer will provide easy-to-use interfaces for farmers and other stakeholders.

- **Mobile Application:** Farmers will use the app to capture images, connect with the density device, and receive results such as yield predictions, disease alerts, market price forecasts, and quality grading reports. The app will also include videos, guides, and tips in local languages.
- **Web Dashboard:** Agriculture officers, buyers, and exporters will have access to a dashboard where they can view reports, quality statistics, and traceability logs. This will help improve transparency and market trust.

3.3 Individual Component

This section focuses on the Post-Harvest Quality Grading Module, which is my individual component of the system. The main goal of this module is to provide farmers with an easy, affordable, and reliable way to evaluate the quality of their pepper after harvesting. At present, grading is mostly done using the traditional cup method or by manual inspection, which can be inaccurate, inconsistent, and sometimes unfair. My proposed solution combines IoT-based bulk density measurement and AI-based visual analysis to introduce a modern grading process that is both farmer-friendly and aligned with international standards.

- **Bulk Density Measurement with IoT**

An IoT device will be developed using an ESP32 microcontroller and a load cell with HX711 amplifier to measure the bulk density of pepper. The device will connect to the mobile application via Bluetooth. Farmers will simply fill a standard cup with pepper, place it on the device, and get a digital reading in grams per liter (g/L). This replaces the traditional manual weighing, which is often error-prone, and ensures consistent and accurate results.

- **Visual Quality Grading with AI**

To check defects, farmers will capture images of pepper samples using their smartphone cameras. They will spread the pepper on a clean A4 sheet, and the app will guide them to take multiple images under good lighting. These images will then be processed using computer vision models such as YOLOv8 (for defect detection) and MobileNetV3 (for texture and surface quality). The analysis will look for issues such as:

- Mold and fungal damage
- Insect bites or discoloration
- Adulteration (e.g., mixing with papaya seeds)
- Size and shape irregularities
- Color uniformity

This approach ensures farmers do not need expensive cameras or lab facilities. A normal smartphone is enough for reliable grading.

- **Piperine Estimation and Indigenous Value**

Since direct chemical testing for piperine is costly and not accessible to smallholder farmers, the system will provide an estimated piperine value based on variety and region. This will emphasize the natural advantage of Ceylon pepper, which is known to have higher piperine levels than many hybrids. The system will encourage farmers to cultivate and preserve indigenous pepper varieties to maintain quality and market value.

- **GAP Certification Validation**

The module will also support GAP validation. If a farmer has a GAP certificate, it can be uploaded and verified within the app. Verified certificates will be linked to the farmer's profile, helping them build trust with buyers and exporters. Farmers without certification will be shown the benefits of GAP and encouraged to apply.

- **Report Generation and Buyer Notifications**

At the end of the process, the system will generate a digital grading report (PDF). This report will summarize:

- Bulk density results
- Visual grading outcomes
- Estimated piperine content
- GAP certification status
- Final grade (Premium, Grade 1, Grade 2, etc.)

Farmers can share this report directly with buyers. If a batch is graded as Premium quality, notifications will be sent to registered buyers and exporters, creating better market opportunities for the farmer.

- **Farmer Guidance and Packaging Instructions**

Along with grading, the system will give practical suggestions to farmers, such as:

- How to avoid mold in future harvests
- Proper drying and sorting techniques
- Standard packaging guidelines with step-by-step images

This ensures that even farmers with little technical knowledge can improve their pepper quality and maintain export standards.

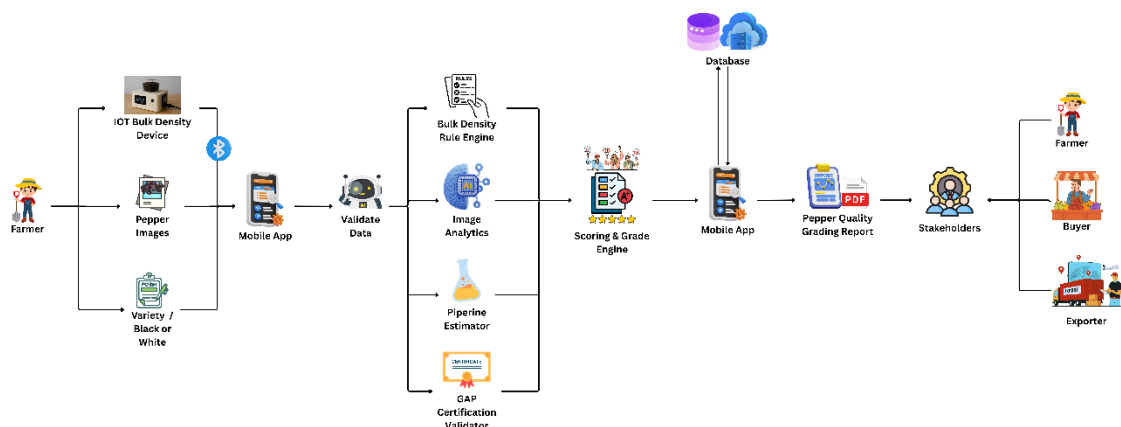


Figure 3.2: System Diagram for Post Harvest Quality Grading

3.4 Software Solution

The proposed software solution is built as a mobile application with AI and IoT support, backed by cloud services for secure storage and reporting. The aim is to provide farmers with an easy-to-use app that can guide them through pepper quality checking, while the heavy AI tasks run in the background.

Mobile Application

- The app will be developed using Flutter, which allows it to run on most Android devices used by farmers.
- It will include features like Bluetooth connection to the density device, step-by-step instructions with videos/images, image capture for quality grading, GAP certificate upload, and instant report viewing.
- The app will work in an offline-first mode, so farmers can still record density and take images even without internet. The data will sync automatically when the connection is back.

AI and Image Processing

- AI models will be trained using TensorFlow/PyTorch and optimized for mobile or cloud inference.
- YOLOv8 will be used to check for color uniformity, defects, mold, and adulteration.
- MobileNetV3 will be used to analyze texture and surface features of pepper samples.
- Results from AI models will be combined with density readings and GAP status to give a final quality grade (Premium, Grade1, Grade2, Grade3).

Backend and Database

- The backend will be developed with FastAPI/Node.js, offering APIs for AI inference, grading logic, and report generation.
- Firebase Firestore will handle user accounts, grading history, and certificate records.
- SQLite will be used inside the app for offline data storage.
- All grading reports will be saved as PDFs in Firebase Cloud Storage, so farmers can download or share them with buyers.

Report and Notifications

- After grading, the system will generate a digital report that shows density, AI image analysis, GAP certificate status, and overall grade.
- Farmers will also receive tips for improvement if defects are found.
- Buyers can get notifications when premium-grade pepper batches are recorded, which supports better market connectivity.

Security and Privacy

- All farmer data (images, certificates, reports) will be encrypted during transfer and storage.
- Farmers will always know how their data is used, and they can delete their records if they want.

3.5 Commercialization

The system holds significant potential for technology transfer and agribusiness impact in Sri Lanka and beyond:

- Target Users
 - Smallholder pepper farmers (majority of Sri Lankan farmers).
 - Plantation cooperatives and spice exporters.
 - Government extension services and NGOs.

- **Business Model**

Free tier: Basic yield prediction by weight.

Premium tier: Advanced AR visualization, historical trend analysis, and detailed heatmaps via subscription.

B2B partnerships: Analytics and traceability solutions for exporters and certification bodies.

- **Value Proposition**

Farmers: Early insights, reduced uncertainty, better resource allocation, increased profitability.

Exporters: Reliable supply chain with predictable volumes.

Government/NGOs: Macro-level crop monitoring and policy planning.

- **Scalability and Adaptability**

The architecture is crop-agnostic and can be extended to other spices (cardamom, cinnamon, nutmeg, etc.).

Potential for regional/global adoption in spice-exporting economies.

3.6 Future Scope

While this research mainly focuses on building an IoT-supported, AI-powered pepper quality grading system, there are several directions for future improvements:

- **Moisture Measurement Integration**

At present, only bulk density is measured. In the future, the IoT device can be extended with a moisture sensor to provide more accurate grading according to international standards.

- **Direct Piperine Testing**

Currently, piperine content is estimated indirectly using variety and region.

Future versions may explore low-cost chemical sensors or spectroscopy methods to directly measure piperine levels on farms.

- **Larger Dataset for AI Training**

As more images are collected from different regions and pepper varieties, the AI models can be retrained to improve accuracy and robustness across lighting conditions, backgrounds, and farmer devices.

- **AI Chatbot and Advisory Support**

A chatbot in Sinhala and Tamil can be added to answer farmers' questions, provide guidance on improving quality, and give quick recommendations for handling defects.

- **Cross-Crop Extension**

The same system can be extended to other spices such as cardamom, cloves, or cinnamon, making it part of a broader smart spice farming ecosystem.

- **Integration with Supply Chain**

Future work can link the grading reports directly with exporters, certification bodies, and blockchain-based traceability systems to strengthen trust in Sri Lankan pepper worldwide.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

Data Collection

- **Pepper Type and Variety Selection:** The farmer will first select whether the pepper is black or white and then choose the specific variety. This ensures that grading results are linked to the right standard [1] - [5].
- **Density Measurement with IoT Device:** The farmer will connect a Bluetooth-enabled density device to the app. The app will provide simple step-by-step guides (with pictures or short videos) to place pepper into the standard cup. The IoT device will then measure the bulk density (g/L). This will reduce errors compared to the traditional 1L cup method [4].
- **Image Capture for Visual Analysis:** Farmers will be guided to spread pepper samples on a clean A4 sheet and capture 8 images. The app will show overlays and prompts (e.g., “Good lighting required,” “Keep steady”) to make sure the photos are clear and usable.
- **GAP Certificate Verification:** The system will check whether the farmer has GAP certification. If their profile already has a verified certificate, it will be validated automatically. If not, the farmer will be asked to upload a copy for admin review [16].

Pre-processing and Analysis

- **Density Data Standardization:** The app will compare measured density against accepted grading thresholds (e.g., ≥ 550 g/L for Black Pepper Grade 1) [1], [3], [4].
- **Image Pre-checks:** Captured images will be checked for clarity, brightness, and blurriness before analysis.
- **Quality Factor Extraction:** Using AI models (YOLOv8, MobileNetV3, etc.), the system will analyze images for:
 - Color uniformity

- Surface texture
- Berry size and shape consistency
- Presence of pinheads
- Extraneous matter (stems, dust, stones)
- Adulteration (e.g., papaya seeds)
- Mold or insect damage [7], [8], [14], [15].
- Rubric-Based Scoring: The system will score the pepper sample by combining bulk density, image-based features, and GAP certification. Variety-based piperine estimation will also be included [9] - [11].

Scoring and Quality Grading

- Composite Score Calculation: The app will calculate a composite score by integrating density, image analysis, and certification results.
- Final Grade Assignment: The pepper will then be graded as Premium, Grade 1, Grade 2, Grade 3, or below, based on configurable thresholds.
- Transparency of Scoring: Farmers will be able to see a rubric showing how points were awarded/deducted, along with tips for improvement.

Visualization and Reporting

- Immediate Feedback: Results from density, image analysis, and final grade will be shown instantly.
- Digital Report Generation: A PDF report will be created with farmer details, batch info, density values, visual analysis, and final grade. Actionable advice will also be included.
- History Tracking: Farmers can access their past grading records to view quality trends over time.

Farmer Guidance and Support

- **Instructional Material:** Each step will have short videos or illustrated guides so farmers can easily follow.
- **Improvement Tips:** If defects (e.g., mold, light berries, adulteration) are detected, the app will give practical advice to reduce these issues in future harvests.
- **Packaging Instructions:** Farmers will receive simple guidance (with steps and images) on packaging their pepper according to export standards.

Buyer and Market Interaction

- **Premium Notifications:** If a farmer's pepper reaches premium grade, registered buyers will be notified. This creates a direct link between quality farmers and markets.
- **Export Readiness Support:** Farmers will be encouraged to get GAP certification and follow proper packaging, making their produce more attractive for export.

Administration and Control

- **Threshold and Rubric Management:** Admins can update grading thresholds, factor weights, and scoring models to stay aligned with ISO/IPC standards [2], [3], [4].
- **Certificate Validation:** Admins will review uploaded GAP certificates and update farmer profiles.
- **Flagged Session Review:** Suspicious cases (e.g., reused images, inconsistent density values) will be flagged for admin checking.

Quality Assurance

- **Data Integrity:** All density values will be linked with device ID and timestamp.
- **Cheating Prevention:** Multi-step photo capture and IoT-based readings will reduce manipulation.
- **Privacy and Consent:** Farmers must give consent before data collection and will have the right to delete their reports or account.

4.2 Non-Functional Requirements

Performance:

The app should provide real-time density readings and process images within about 1 minute on normal farmer devices. It should also allow offline use and sync data when internet is available.

Accuracy:

The IoT device must give consistent bulk density readings through calibration. The AI models should detect key defects (mold, adulteration, light berries) with acceptable accuracy, in line with ISO/IPC standards [1] - [4].

Usability:

The system must be simple for farmers with limited technical knowledge. Clear step-by-step guides (videos, images) will be provided.

Security & Privacy:

Farmer data, images, and certificates must be stored securely with encryption. Only authorized users can access reports. Farmers will have control over their data and the option to delete it.

Reliability:

The app should work smoothly on low-cost Android devices. IoT device connections must be stable, and no data should be lost during poor connectivity.

Scalability & Adaptability:

The system should support more farmers, pepper varieties, and buyers over time. Grading thresholds, rubrics, and packaging instructions should be configurable by admins to match new standards.

4.3 User Requirements

For Farmers

- **Ease of Use:**
The app must be simple for farmers with minimal technical knowledge. Step-by-step guides, images, and short videos should be provided.
- **Guided Instructions:**
Farmers should get clear guidance when connecting the density device, capturing images, and uploading GAP certificates.
- **Real-time Feedback:**
Farmers should see density readings instantly and get quick feedback on pepper quality.
- **Transparency of Grading:**
Farmers must be able to see how the grading was done. A simple rubric should show why points were added or reduced.
- **Access to Past Results:**
Farmers should be able to view their history of quality checks, spot trends, and learn from past mistakes.
- **Improvement Tips:**
If defects (mold, adulteration, or poor drying) are found, the app should give practical suggestions to improve the next harvest.
- **Packaging Guidance:**
Farmers should get easy-to-follow steps (with images) on how to package pepper properly for export standards.
- **Data Privacy:**
Farmers should know that their data is safe, used only for grading and market purposes, and can be deleted if they request.

- **Offline Support:**
Since many farmers work in low-connectivity areas, the app should work offline and sync later when internet is available.

For Buyers

- **Premium Quality Alerts:**
Buyers should receive notifications when farmers produce premium-grade pepper, making it easier to source high-quality batches.
- **Trust and Transparency:**
Buyers should be able to view digital grading reports, certification status, and farmer details before making a purchase.

For Admins

- **Grading Management:**
Admins should be able to update grading thresholds, scoring rubrics, and factor weights according to international standards.
- **Certificate Validation:**
Admins must be able to check and approve GAP certificates uploaded by farmers.
- **Fraud Prevention:**
The system should allow admins to review flagged sessions (e.g., reused images or inconsistent density values).

4.4 System Requirements

Hardware Requirements

For Farmers (End Users):

- Smartphone (Android, minimum 3 GB RAM, Android 9 or above).
- Built-in camera ($\geq$ 12MP resolution for clear sample images).

- Bluetooth connectivity (for IoT density device).
- Internet access (Wi-Fi/4G) for syncing data and generating reports.

For IoT Density Device:

- ESP32 microcontroller with Bluetooth support.
- Load cell with HX711 amplifier for bulk density measurement.
- Standard density cup for consistent measurements.
- Rechargeable battery ($\geq 2000\text{mAh}$) for field use.

For Developers / Research Team:

- Processor: Intel i5/i7 (or equivalent).
- GPU: NVIDIA RTX 2060/3060 (or higher) for image model training.
- RAM: Minimum 16 GB.
- Storage: 1 TB SSD for datasets, checkpoints, and reports.

For Server / Cloud Deployment (optional, if scaling):

- Cloud server or VPS with minimum 8-core CPU, 32 GB RAM.
- Storage with automated backup (≥ 2 TB).

Software Requirements

For Mobile Application Development:

- IDE: Android Studio / Visual Studio Code.
- Framework: Flutter SDK (cross-platform).
- Language: Dart (for app) and Java/Kotlin (for device drivers if required).
- Version Control: Git & GitHub for collaborative development.

For Backend and API Services:

- Node.js / FastAPI for backend services and AI inference.
- Firebase Authentication & Firestore for user accounts, storage, and syncing.

- SQLite for offline data storage on farmer devices.

For AI/ML Model Training:

- Python (3.8 or above).
- TensorFlow / PyTorch for training computer vision models.
- OpenCV for image preprocessing and feature extraction.
- Jupyter Notebook for experimentation and model evaluation.

For Report Generation:

- Python ReportLab / FPDF library for generating PDF reports.
- Cloudinary Storage for storing reports securely.

For Testing and Deployment:

- Flutter Testing Framework for unit and integration testing.
- Docker (optional) for backend containerization.

Computing Resources

- Local GPU-enabled machines for AI model training and dataset preprocessing.
- Cloud computing resources (Google Cloud / AWS / Azure) for large-scale training or when datasets exceed local hardware capacity.
- Scheduled batch processing for model updates to avoid slowing down real-time farmer usage.

Networking Requirements

- Stable internet (3G/4G or Wi-Fi) for syncing images, certificates, and reports.
- Offline-first design: farmers can complete data collection offline and sync later.
- Bluetooth (BLE) connectivity between IoT density device and mobile app.
- Secure HTTPS communication for all server and database interactions.

5. GANTT CHART



Figure 5.1: Gantt Chart

6. WORK BREAKDOWN STRUCTURE

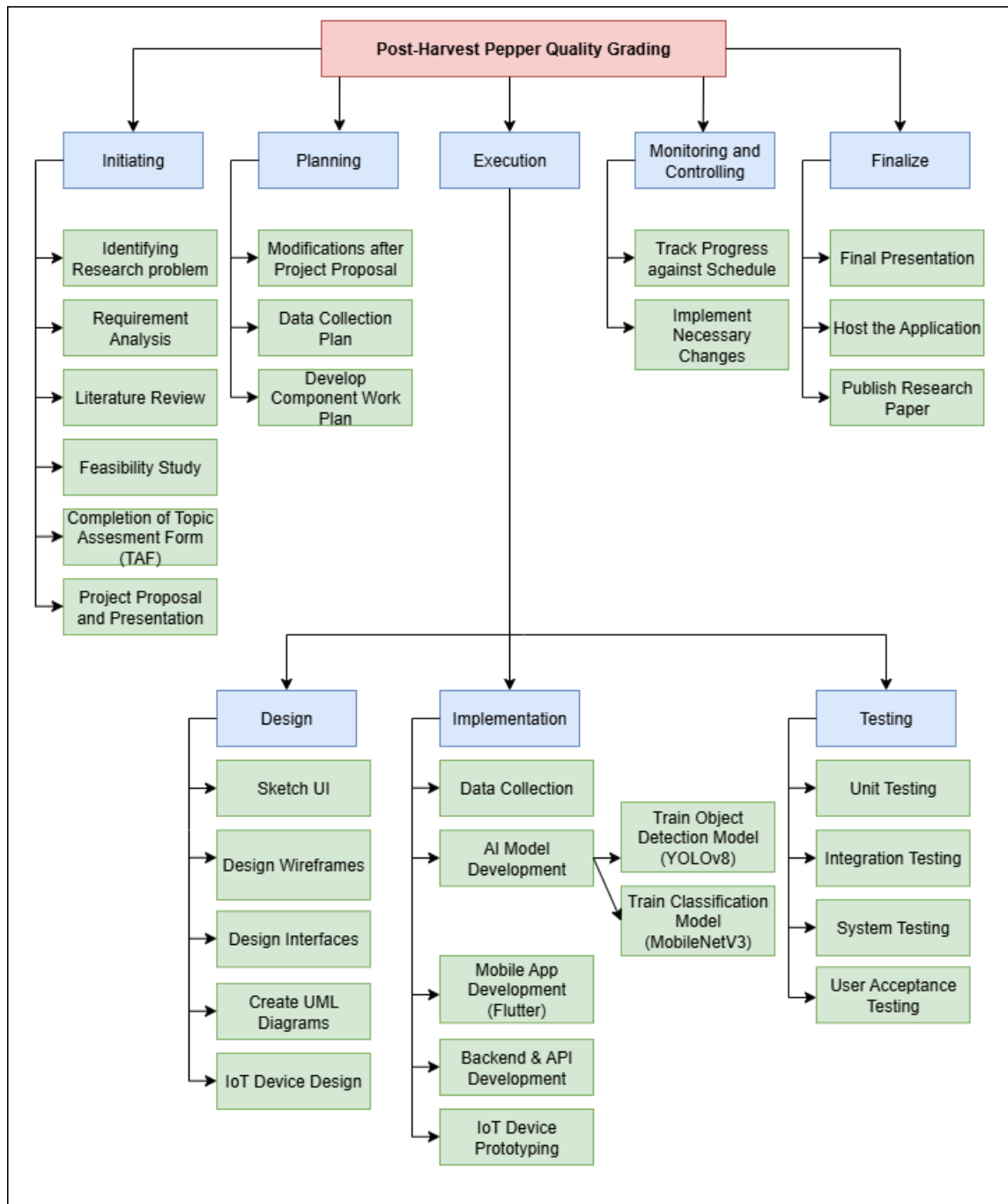


Figure 6.1: Work Breakdown Structure

7. BUDGET AND BUDGET JUSTIFICATION

Estimated Budget for IoT Device Prototype

Table 7.1: Budget for IoT Device

Requirement	Quantity	Unit Cost (Rs.)	Total (Rs.)
ESP32-WROOM	1	1150	1150
Load Cell (1kg)	1	480	480
HX711 Amplifier	1	150	150
Power Supply	1	540	540
1L measuring cup	1	895	895
Buck Converter	1	270	270
Jumper wires	1	150	150
Enclosure box	1	2500	2500
AC wires (2m)	1	425	425
Plug top	1	250	250
Push button	1	60	60
LED indicator	1	170	170
Total Cost			7040

8. DESCRIPTION OF PERSONNEL AND FACILITIES

Ms. Ishara Weerathunga (Supervisor)- Sri Lanka Institute of Information Technology (SLIIT).

Mr. Eishan Weerasinghe (Co-Supervisor)- Sri Lanka Institute of Information Technology (SLIIT).

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APPENDICES

Appendix A - Plagiarism Report

The screenshot displays a Turnitin Feedback Studio interface. The main document area shows the title "AI BASED INTEGRATED SOLUTION FOR SMART PEPPER FARMING IN SRI LANKA", the ID "25-26J-479", and the author "Adhikari A. A. J. T.". The right sidebar features a "Match Overview" section with a 3% similarity score. Below this, a list of matches is shown, including "Submitted to National I...", "Submitted to Sri Lanka...", "Nilini A. Wimalaratna...", "rspaciencehub.com", and "B. Sasikumar, Santhos...". The bottom status bar indicates "Page: 1 of 40" and "Word Count: 7529".

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